



User Management in Linux

Linux is a **multi-user operating system**, meaning multiple users can access and work on the same system simultaneously.

User management involves:

- Creating users
- Modifying users
- Deleting users
- Managing passwords
- Controlling login access
- Monitoring users

(🚫 We will NOT cover group management here.)

1 Types of Users in Linux

1. Root User

- Username: `root`
- UID: `0`
- Has complete control over the system
- Home directory: `/root`
- Can access and modify any file

Check root:

`id root`

2. Normal (Regular) Users

- Used for daily work
- Limited permissions
- Home directory: `/home/username`
- UID usually starts from 1000

Example:

`/home/ashvi`

3. System / Service Users

- Created automatically for services
- Usually no login shell
- UID range: 1–999 (varies by distro)

Example users:

- `daemon`
 - `nobody`
 - `apache`
-

2 Important User-Related Files

/etc/passwd

Stores basic user account information.

View:

```
cat /etc/passwd
```

Example entry:

```
ashvi:x:1001:1001:Ashvi:/home/ashvi:/bin/bash
```

Field explanation:

```
username : password_placeholder : UID : GID : comment : home_dir :  
shell
```

Important:

- Password field shows `x`
 - Actual encrypted password stored in `/etc/shadow`
-

/etc/shadow

Stores hashed passwords.

Only root can read it:

```
sudo cat /etc/shadow
```

Contains:

- Hashed password
- Password expiry info
- Account expiry info

Format:

username:hashed_password:last_change:min:max:warn:inactive:expire:reserved

Field Explanation

Field	Meaning
username	Whose account
hashed_password	Secret scrambled password
last_change	Days since Jan 1, 1970 when password last changed
min	Minimum days before password can be changed
max	Maximum days password is valid
warn	Days before expiry to warn user
inactive	Grace period after expiry
expire	Account expiration date

reserved

Reserved

Important Things to Remember

- Passwords are **hashed**, not encrypted.
- Only root can see this file.
- If the password field has **!** → account locked.
- If password field has ***** → login disabled.

Simple Comparison

Symbol	Meaning	What Happens
!	Locked	Password was valid, but now blocked
*	Disabled	No password login possible at all

/etc/login.defs

The **UID (User ID)** and **GID (Group ID)** range information is stored in:

What These Mean

Variable	Purpose
UID_MIN	Minimum UID for normal users

<code>UID_MAX</code>	Maximum UID for normal users
<code>GID_MIN</code>	Minimum GID for normal groups
<code>GID_MAX</code>	Maximum GID for normal groups
<code>SYS_UID_MIN/</code> <code>MAX</code>	Range for system users
<code>SYS_GID_MIN/</code> <code>MAX</code>	Range for system groups

Typical Linux UID Layout

UID Range		Type
0	root	
1–999	System users (services like apache, nginx, mysql)	
1000+	Regular users	

Creating Users

There are two main commands:

◆ **useradd (Basic Method)**

`sudo useradd username`

 By default:

- May NOT create home directory

- No password set
 - Cannot login until password is set
-

◆ **useradd with Options**

```
sudo useradd -m -s /bin/bash ashvi
```

Options:

- **-m** → Create home directory
- **-s** → Specify shell
- **-c** → Add comment

Example:

```
sudo useradd -m -s /bin/bash -c "DevOps Engineer" ashvi
```

◆ **adduser (Recommended – Interactive)**

```
sudo adduser ashvi
```

- ✓ Creates home directory
- ✓ Prompts for password
- ✓ Asks for user details

More user-friendly (Debian/Ubuntu systems).

4 Setting and Changing Password

Set password for user:

```
sudo passwd ashvi
```

Change your own password:

```
passwd
```

Force password change on next login:

```
sudo passwd -e ashvi
```

5 Viewing User Information

Check current user:

```
whoami
```

Check user ID:

```
id ashvi
```

Output example:

```
uid=1001(ashvi)
```

List logged-in users:

```
who
```

```
w
```

6 Switching Users

Switch to another user:

```
su ashvi
```

Switch to root:

```
su -
```

Switch using sudo:

```
sudo -i
```

7 Modifying Users (usermod)

Change username:

```
sudo usermod -l newname oldname
```

Change home directory:

```
sudo usermod -d /home/newdir -m ashvi
```

`-m` moves existing files.

Change login shell:

```
sudo usermod -s /bin/zsh ashvi
```

Common shells:

- `/bin/bash`
 - `/bin/sh`
 - `/bin/zsh`
 - `/sbin/nologin` (disable login)
-

8 Locking and Unlocking Users

Lock user account:

```
sudo usermod -L ashvi
```

OR

```
sudo passwd -l ashvi
```

Unlock:

```
sudo usermod -U ashvi
```

9 Deleting Users

Delete user only:

```
sudo userdel ashvi
```

Delete user + home directory:

```
sudo userdel -r ashvi
```

⚠ Always backup important data before deleting.

10 Disabling User Login

Disable shell:

```
sudo usermod -s /sbin/nologin ashvi
```

OR

```
sudo usermod -s /bin/false ashvi
```

Users cannot log in.

+ Including chage (Password Aging) & Default User Creation Settings

This guide focuses only on **user management** (no group management), with special emphasis on:

- ✓ **chage** command (password aging control)
 - ✓ Default user creation settings
 - ✓ Account expiry & security policies
-

1 Understanding User Account Lifecycle

When a user is created in Linux:

1. Entry is added to `/etc/passwd`
 2. Password is stored in `/etc/shadow`
 3. Home directory is created (if `-m` used)
 4. Default settings are applied from:
 - `/etc/login.defs`
 - `/etc/default/useradd`
 - `/etc/skel`
-

2 Password Aging & Expiry – `chage` Command (Very Important 🔥)

👉 What is `chage`?

`chage` = Change Age

It is used to manage:

- Password expiry
 - Account expiry
 - Password warning days
 - Minimum & maximum password validity
-

◆ Check Current Password Aging Information

```
chage -l username
```

Example:

```
chage -l ashvi
```

Example output:

Last password change	: Feb 16, 2026
Password expires	: Mar 18, 2026
Password inactive	: never
Account expires	: never
Minimum number of days between password change	: 0
Maximum number of days between password change	: 30
Number of days of warning before password expires	: 7

3 Understanding **chage** Parameters

◆ 1. Set Maximum Password Validity

```
sudo chage -M 30 username
```

Meaning:

- Password will expire after 30 days.
-

◆ 2. Set Minimum Days Between Password Changes

```
sudo chage -m 7 username
```

User cannot change password before 7 days.

◆ 3. Set Warning Days Before Expiry

```
sudo chage -W 5 username
```

User will get warning 5 days before password expires.

◆ 4. Set Account Expiry Date

```
sudo chage -E 2026-12-31 username
```

After that date:

- User cannot login.
-

◆ 5. Force Password Change at Next Login

```
sudo chage -d 0 username
```

OR

```
sudo passwd -e username
```

User must change password immediately.

◆ 6. Disable Account Expiry

```
sudo chage -E -1 username
```

Removes account expiration.

◆ 7. Change All at Once (Interactive Mode)

```
sudo chage username
```

It will ask:

Minimum Password Age

Maximum Password Age

Password Expiration Warning

Account Expiration Date

4 Complete **chage** Example (Real Scenario)

Set company policy:

- Password expires in 60 days
- Minimum 7 days before change
- Warning 10 days before expiry
- Account expires on 31 Dec 2026

```
sudo chage -M 60 -m 7 -W 10 -E 2026-12-31 ashvi
```

Verify:

```
chage -l ashvi
```

5 What Happens When Passwords Expire?

- User gets warning
 - After expiry → login denied
 - Admin must reset password
-

6 Difference: Account Expiry vs Password Expiry

Feature	Password Expiry	Account Expiry
Controlled by	-M option	-E option
Effect	Must change password	Cannot login at all
Common use	Security policy	Temporary employees

7 Default User Creation Settings (Very Important 🔥)

When you create a user, Linux uses default settings from these files:

1. `/etc/login.defs`

Controls:

- Default UID range

- Password aging defaults
- Home directory creation settings

Check:

```
cat /etc/login.defs
```

Important parameters:

◆ UID Range

```
UID_MIN 1000
UID_MAX 60000
```

◆ Password Aging Defaults

```
PASS_MAX_DAYS 99999
PASS_MIN_DAYS 0
PASS_WARN_AGE 7
```

Meaning:

- Password valid for 99999 days
- No minimum days
- 7-day warning

1 Set Default Password Aging Policy (For All NEW Users)

If you want to enforce the **same password policy for all users**, then this is controlled by:
/etc/login.defs

1. **Edit it: `sudo vi /etc/login.defs`**
2. Look for these parameters:

PASS_MAX_DAYS 90

PASS_MIN_DAYS 7

PASS_WARN_AGE 7

3. Meaning:

Setting	Effect
PASS_MAX_DAYS 90	Password expires in 90 days
PASS_MIN_DAYS 7	User must wait 7 days before changing
PASS_WARN_AGE 7	Warn 7 days before expiry

⚠ Important:

This applies only to **new users created after the change.**

2 Apply Policy to Existing Users

1. To enforce for all existing users:

Example: `sudo chage -M 90 -m 7 -W 7 username`

2. To apply for all normal users (UID ≥ 1000):

for user in \$(awk -F: ' \$3 >= 1000 {print \$1}' /etc/passwd); do

`sudo chage -M 90 -m 7 -W 7 $user`

done

Now all users follow the same aging policy.

3 Enforce Password Complexity (Strong Password Rules)

Password length, special characters, etc. are controlled by PAM.

Modern Linux uses: `/etc/security/pwquality.conf`

1. **Edit: `sudo vi /etc/security/pwquality.conf`**
2. **Example strong policy:**

`minlen = 12`

`dcredit = -1`

`ucredit = -1`

`lcredit = -1`

`ocredit = -1`

`maxrepeat = 3`

3. **Meaning:**

Setting	Meaning
<code>minlen=12</code>	Minimum 12 characters
<code>dcredit=-1</code>	At least 1 digit
<code>ucredit=-1</code>	At least 1 uppercase
<code>lcredit=-1</code>	At least 1 lowercase
<code>ocredit=-1</code>	At least 1 special character
<code>maxrepeat=3</code>	No more than 3 repeated characters

4 Prevent Password Reuse

1. Edit PAM config:

On RHEL/CentOS: /etc/pam.d/system-auth

On Ubuntu: /etc/pam.d/common-password

2. Add or ensure:

`password required pam_pwhistory.so remember=5`

3. Meaning:

User cannot reuse last 5 passwords.

5 Lock Account After Failed Attempts

1. Edit: `/etc/pam.d/system-auth`
2. Add:

`auth required pam_faillock.so preauth silent deny=5 unlock_time=900`

`auth required pam_faillock.so authfail deny=5 unlock_time=900`

3. Meaning:
 - Lock after 5 failed attempts
 - Unlock after 15 minutes (900 seconds)
-

2. `/etc/default/useradd`

Contains default settings for `useradd`.

Check:

```
cat /etc/default/useradd
```

Example:

```
GROUP=100  
HOME=/home  
INACTIVE=-1  
EXPIRE=  
SHELL=/bin/bash  
SKEL=/etc/skel  
CREATE_MAIL_SPOOL=yes
```

Field-by-Field Explanation

Parameter	Meaning
GROUP=100	Default primary group ID assigned to new users
HOME=/home	Base directory where user home directories are created
INACTIVE=-1	Days after password expiry before account is disabled
EXPIRE=	Account expiration date (empty = never expires)
SHELL=/bin/bash	Default login shell
SKEL=/etc/skel	Skeleton directory copied into new user's home
CREATE_MAIL_SPOOL=yes	Whether to create mail file in <code>/var/mail/username</code>

3. `/etc/skel`

This directory contains default files copied to new users.

Check:

```
ls -a /etc/skel
```

Common files:

```
.bashrc  
.bash_profile  
.profile
```

When new user is created:

```
sudo useradd -m ashvi
```

Contents of `/etc/skel` are copied into:

```
/home/ashvi
```

8 Changing Default Settings for New Users

◆ Change Default Shell

```
sudo useradd -D -s /bin/zsh
```

Check:

```
useradd -D
```

◆ Change Default Expiry

```
sudo useradd -D -e 2026-12-31
```

◆ Change Default Inactive Days

```
sudo useradd -D -f 30
```

Meaning:

Account disabled 30 days after password expiry.

9 Checking Default Settings

```
useradd -D
```

Example output:

```
GROUP=100  
HOME=/home  
INACTIVE=-1  
EXPIRE=  
SHELL=/bin/bash
```

10 Security Best Practices

- ✓ Set password expiry (60–90 days)
 - ✓ Set minimum password age
 - ✓ Set warning days
 - ✓ Disable inactive accounts
 - ✓ Use account expiry for temporary users
 - ✓ Regularly check `chage -l`
-



Quick Command Summary

Command	Purpose
<code>chage -l</code>	Check aging
<code>chage -M</code>	Set max days
<code>chage -m</code>	Set min days

chage -W	Warning days
chage -E	Account expiry
chage -d 0	Force password change
useradd -D	Show defaults
useradd -D -s	Change default shell
